

A Note on Theorems 1 and 3 of Zhou and Liang (2018)

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1 Introduction and Theorem Statements

[Zhou and Liang, 2018] characterize the critical points of the squared error loss for a deep linear network:

$$L(A_1, \dots, A_\ell) = \frac{1}{2} \|A_\ell \cdots A_2 A_1 X - Y\|_F^2.$$

In this note, we first discuss a technical issue in the proof of Theorem 1, and recursively in the proof of Theorem 3. We then provide a repair of their result, preserving the validity of their parametrization of critical points. Both the issue in the original proof and the validity of our alternative argument were confirmed by the authors in private correspondence.

In Theorem 1, they consider the case of one hidden layer:

$$\mathcal{L}(A_1, A_2) = \frac{1}{2} \|A_2 A_1 X - Y\|_F^2.$$

Under their parametrization

$$A_2 = UVC,$$

with C invertible and $U, \widehat{V}$ having orthonormal columns, where $V = [\widehat{V}, 0]$, they obtain the solution for $\nabla_{A_1} \mathcal{L} = 0$:

$$A_1 = C^{-1} V^\top U^\top Y X^\dagger + L_1 - C^{-1} V^\top V C L_1 X X^\dagger.$$

Theorem 3 extends the two-layer characterization recursively to deep linear networks. Using the notation

$$A_{(\ell,k+1)} := A_\ell \cdots A_{k+1}, \quad A_{(k-1,1)} := A_{k-1} \cdots A_1, \quad A_{(0,1)} = I,$$

their equation (4) parametrizes the solutions to

$$\nabla_{A_k} \mathcal{L}_D = 0$$

for each intermediate layer A_k , given the recursively determined value of $A_{(\ell,k+1)}$.

2 A Counterexample

The proof of Theorem 1 in [Zhou and Liang, 2018] applies a pseudoinverse identity that does not hold in general. Specifically, after showing that

$$A_2 = UVC,$$

the proof substitutes

$$A_2^\dagger = C^{-1}V^\top U^\top.$$

However, this equality does not hold in general given the definitions of C, V, U .

For example, let

$$U = \begin{bmatrix} 1 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then

$$A_2 = UVC = \begin{bmatrix} 1 & 1 \end{bmatrix},$$

so

$$A_2^\dagger = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \quad C^{-1}V^\top U^\top = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Thus $A_2^\dagger \neq C^{-1}V^\top U^\top$ in general.

The same argument is followed recursively in Theorem 3. We show below that their parametrizations from Theorems 1 and 3 are still valid by bypassing this pseudoinverse step with an alternate proof based on generalized inverses.

3 Reference Identities and Properties

We will repeatedly use Theorem 2.3.2 of Rao and Mitra for the general solutions to matrix equations [Rao and Mitra, 1971]. In all applications considered, the equations are consistent, so we will directly apply the solution formulas.

The Moore-Penrose pseudoinverse of W is denoted by $W^\dagger$.

Theorem 3.1 (General Case). *Let A, B, C be matrices of compatible dimensions, and let A^- and B^- be generalized inverses of A and B , respectively; that is, $AA^-A = A$ and $BB^-B = B$. If the matrix equation*

$$AWB = C$$

is consistent, then its general solution is

$$W = A^-CB^- + Z - A^-AZBB^-,$$

where Z is an arbitrary matrix of compatible dimensions.

Property 3.2 (Properties of the Moore-Penrose Pseudoinverse). *Let $P \in \mathbb{R}^{d \times n}$. The Moore-Penrose pseudoinverse $P^\dagger$ satisfies:*

- (i) $P^\top(PP^\top)^\dagger = P^\dagger$.
- (ii) $PP^\dagger P = P$.
- (iii) $P^\dagger PP^\top = P^\top$.
- (iv) $P^\top PP^\dagger = P^\top$.

4 Correction to Theorem 1

Recall that a generalized inverse of a matrix A is a matrix A^- such that $AA^-A = A$.

Lemma 4.1. *Define invertible matrix C and matrices with orthonormal columns $U, \widehat{V}$, and let $V = [\widehat{V}, 0]$. Let $G = C^{-1}V^\top U^\top$ and $A_2 = UVC$. Then $GG^\top$ is a generalized inverse of $A_2^\top A_2$. Also, $(XX^\top)^\dagger$ is a generalized inverse of $XX^\top$.*

Proof. Observe that

$$\begin{aligned} GG^\top &= C^{-1}V^\top U^\top (C^{-1}V^\top U^\top)^\top \\ &= C^{-1}V^\top U^\top UVC^{-\top} \\ &= C^{-1}V^\top VC^{-\top} \end{aligned}$$

and

$$\begin{aligned} A_2^\top A_2 &= C^\top V^\top U^\top UVC \\ &= C^\top V^\top VC. \end{aligned}$$

Hence

$$\begin{aligned} A_2^\top A_2 (GG^\top) A_2^\top A_2 &= (C^\top V^\top VC) (C^{-1}V^\top VC^{-\top}) (C^\top V^\top VC) \\ &= C^\top (V^\top V)^3 C. \end{aligned}$$

Now since $V = [\widehat{V}, 0]$, we have

$$V^\top V = \begin{bmatrix} \widehat{V}^\top \\ 0 \end{bmatrix} \begin{bmatrix} \widehat{V} & 0 \end{bmatrix} = \begin{bmatrix} \widehat{V}^\top \widehat{V} & 0 \\ 0 & 0 \end{bmatrix}.$$

But since $\widehat{V}$ has orthonormal columns, $\widehat{V}^\top \widehat{V} = I$, so

$$V^\top V = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}.$$

Hence $(V^\top V)^2 = V^\top V$, so $(V^\top V)^3 = V^\top V$. Thus

$$\begin{aligned} A_2^\top A_2 (GG^\top) A_2^\top A_2 &= C^\top (V^\top V)^3 C \\ &= C^\top (V^\top V) C \\ &= A_2^\top A_2, \end{aligned}$$

so $GG^\top$ is a generalized inverse of $A_2^\top A_2$.

Finally, by Property 3.2(ii),

$$(XX^\top)(XX^\top)^\dagger(XX^\top) = XX^\top,$$

so $(XX^\top)^\dagger$ is a generalized inverse of $XX^\top$. □

Proposition 4.2. *Let*

$$\mathcal{L}(A_1, A_2) = \frac{1}{2} \|A_2 A_1 X - Y\|_F^2.$$

*For invertible matrix C and matrices with orthonormal columns $U, \widehat{V}$, and $V = [\widehat{V}, 0]$, $A_2 = UVC$ and $G = C^{-1}V^\top U^\top$,
the set*

$$\{GYX^\dagger + L_1 - GA_2 L_1 X X^\dagger : L_1 \in \mathbb{R}^{d_1 \times d_0}\}$$

is a necessary-and-sufficient solution set to $\nabla_{A_1} \mathcal{L} = A_2^\top (A_2 A_1 X - Y) X^\top = 0$.

Proof. First we show that the gradient equation is consistent. Take

$$A_1^{(0)} = A_2^\dagger Y X^\dagger.$$

Then

$$\begin{aligned} A_2^\top (A_2 A_1^{(0)} X - Y) X^\top &= A_2^\top (A_2 (A_2^\dagger Y X^\dagger) X - Y) X^\top \\ &= (A_2^\top A_2 A_2^\dagger) Y (X^\dagger X X^\top) - A_2^\top Y X^\top. \end{aligned}$$

Now by Property 3.2(iii), (iv), we have

$$\begin{aligned} (A_2^\top A_2 A_2^\dagger) Y (X^\dagger X X^\top) - A_2^\top Y X^\top &= A_2^\top Y X^\top - A_2^\top Y X^\top \\ &= 0, \end{aligned}$$

so the equation is consistent.

Expanding and grouping terms,

$$\begin{aligned} A_2^\top A_2 A_1 X X^\top - A_2^\top Y X^\top &= 0 \\ (A_2^\top A_2) A_1 (X X^\top) &= A_2^\top Y X^\top. \end{aligned}$$

Thus we may apply Theorem 3.1 to this equation, with $(A_2^\top A_2)^- = GG^\top$ and $(X X^\top)^- = (X X^\top)^\dagger$. We obtain the general solution set

$$A_1 = (GG^\top)(A_2^\top Y X^\top)(X X^\top)^\dagger + L_1 - (GG^\top)(A_2^\top A_2)L_1(X X^\top)(X X^\top)^\dagger,$$

for arbitrary matrix L_1 of appropriate dimensions.

Now,

$$\begin{aligned} GG^\top A_2^\top &= C^{-1}V^\top V C^{-\top} C^\top V^\top U^\top \\ &= C^{-1}V^\top V V^\top U^\top, \end{aligned}$$

and since

$$V^\top V = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix},$$

we have

$$V^\top V V^\top = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \widehat{V}^\top \\ 0 \end{bmatrix} = \begin{bmatrix} \widehat{V}^\top \\ 0 \end{bmatrix} = V^\top.$$

Hence

$$G G^\top A_2^\top = C^{-1} V^\top U^\top = G.$$

Next, by Property 3.2(i),

$$X^\top (X X^\top)^\dagger = X^\dagger,$$

so making these substitutions, our solution set becomes

$$A_1 = G Y X^\dagger + L_1 - G A_2 L_1 X X^\dagger.$$

□

Since

$$G = C^{-1} V^\top U^\top,$$

we have

$$G A_2 = C^{-1} V^\top U^\top U V C = C^{-1} V^\top V C,$$

yielding final solution set

$$A_1 = C^{-1} V^\top U^\top Y X^\dagger + L_1 - C^{-1} V^\top V C L_1 X X^\dagger.$$

This is precisely equation (1) of [Zhou and Liang, 2018].

5 Correction to Theorem 3

Corollary 5.1. *Let*

$$\mathcal{L}_D(A_1, \dots, A_\ell) = \frac{1}{2} \|A_\ell \cdots A_2 A_1 X - Y\|_F^2,$$

and for $k = 1, \dots, \ell - 1$, define

$$A_{(\ell, k+1)} := A_\ell A_{\ell-1} \cdots A_{k+1}, \quad A_{(k-1, 1)} := A_{k-1} \cdots A_1,$$

with $A_{(0,1)} = I$. Suppose that for an invertible matrix C_{k-1} , and matrices U_{k-1} and $\widehat{V}_{k-1}$ with orthonormal columns, and $V_{k-1} = [\widehat{V}_{k-1}, 0]$, we have

$$A_{(\ell,k+1)} = U_{k-1}V_{k-1}C_{k-1}.$$

Define

$$G_{k-1} := C_{k-1}^{-1}V_{k-1}^\top U_{k-1}^\top.$$

Then the set

$$\{G_{k-1}Y(A_{(k-1,1)}X)^\dagger + L_k - G_{k-1}A_{(\ell,k+1)}L_kA_{(k-1,1)}X(A_{(k-1,1)}X)^\dagger : L_k \in \mathbb{R}^{d_k \times d_{k-1}}\}$$

is a necessary-and-sufficient solution set to

$$\nabla_{A_k} \mathcal{L}_D = A_{(\ell,k+1)}^\top (A_{(\ell,k+1)}A_kA_{(k-1,1)}X - Y) (A_{(k-1,1)}X)^\top = 0.$$

Equivalently, since

$$G_{k-1}A_{(\ell,k+1)} = C_{k-1}^{-1}V_{k-1}^\top V_{k-1}C_{k-1},$$

the complete solution set is

$$A_k = C_{k-1}^{-1}V_{k-1}^\top U_{k-1}^\top Y(A_{(k-1,1)}X)^\dagger + L_k - C_{k-1}^{-1}V_{k-1}^\top V_{k-1}C_{k-1}L_kA_{(k-1,1)}X(A_{(k-1,1)}X)^\dagger.$$

Thus, replacing k by $k+1$, this is exactly equation (4) of Theorem 3 in [Zhou and Liang, 2018].

Proof. Fix $k \in \{1, \dots, \ell-1\}$. Zhou and Liang derive the gradient equation

$$\nabla_{A_k} \mathcal{L}_D = A_{(\ell,k+1)}^\top (A_{(\ell,k+1)}A_kA_{(k-1,1)}X - Y) (A_{(k-1,1)}X)^\top = 0$$

Apply Proposition 4.2 with

$$A_2 = A_{(\ell,k+1)} \quad A_1 = A_k \quad X = A_{(k-1,1)}X.$$

By assumption,

$$A_{(\ell,k+1)} = U_{k-1}V_{k-1}C_{k-1},$$

where C_{k-1} is invertible, U_{k-1} and $\widehat{V}_{k-1}$ have orthonormal columns, and $V_{k-1} = [\widehat{V}_{k-1}, 0]$. We similarly define

$$G_{k-1} := C_{k-1}^{-1}V_{k-1}^\top U_{k-1}^\top.$$

Then the complete solution set for A_k is

$$A_k = G_{k-1}Y(A_{(k-1,1)}X)^\dagger + L_k - G_{k-1}A_{(\ell,k+1)}L_kA_{(k-1,1)}X(A_{(k-1,1)}X)^\dagger.$$

As before,

$$\begin{aligned} G_{k-1}A_{(\ell,k+1)} &= C_{k-1}^{-1}V_{k-1}^\top U_{k-1}^\top U_{k-1}V_{k-1}C_{k-1} \\ &= C_{k-1}^{-1}V_{k-1}^\top V_{k-1}C_{k-1}. \end{aligned}$$

Hence

$$A_k = C_{k-1}^{-1}V_{k-1}^\top U_{k-1}^\top Y(A_{(k-1,1)}X)^\dagger + L_k - C_{k-1}^{-1}V_{k-1}^\top V_{k-1}C_{k-1}L_kA_{(k-1,1)}X(A_{(k-1,1)}X)^\dagger.$$

Replacing k by $k+1$ gives exactly equation (4) of Theorem 3 in [Zhou and Liang, 2018].

□

References

- [Rao and Mitra, 1971] Rao, C. R. and Mitra, S. K. (1971). *Generalized Inverse of Matrices and its Applications*. John Wiley & Sons, New York.
- [Zhou and Liang, 2018] Zhou, Y. and Liang, Y. (2018). Critical points of linear neural networks: Analytical forms and landscape properties. In *International Conference on Learning Representations*.